Affective Susceptibility of Investors: A Pilot Study of Stock Return Co-Movement with Emotional Media Content

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***Abstract*—We investigate whether aggregated media sentiment exhibits directional influence consistent with short-horizon stock return variation in a financial decision environment. Using 130,000+ media articles across five U.S. equities (Apple, Alphabet, Amazon, Nvidia, and Tesla) from March 1 to May 4, 2025, we compute daily sentiment using VADER and examine its relationship with daily log returns. To move beyond contemporaneous correlation, we incorporate three complementary analyses: same-day association, lagged sentiment-to-return tests, and market-controlled regressions using SPY as a broad-market proxy. We additionally examine whether extreme sentiment realizations are followed by distinguishable next-day return behavior. The five case studies reveal heterogeneous patterns: Apple, Alphabet, Amazon, and Nvidia exhibit moderate positive association during the study window, whereas Tesla exhibits near-zero association despite substantially greater media volume. We interpret these findings through a behavioral lens that includes investor attention, media saturation, and noise discounting. Although the study does not claim definitive causal identification, it provides directional evidence consistent with sentiment-linked return effects under controlled conditions. The findings motivate future work using broader samples, longer horizons, richer controls, and stronger identification strategies.**

***Index Terms*—affective computing, investor sentiment, media sentiment, stock returns, behavioral finance, financial text analysis, cognitive security**

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I. Introduction

Financial markets respond not only to firm fundamentals and macroeconomic conditions, but also to the information environment in which investors operate [15]. News headlines, opinion pieces, and other digitally distributed narratives can shape expectations, amplify investor attention, and contribute to short-horizon trading behavior [6], [16], [17]. Prior research demonstrates that media tone and public sentiment extracted from textual data can influence investor perceptions and market activity, particularly when information spreads rapidly through digital platforms [21], [14]. This broader informational environment is relevant to affective computing because emotionally charged content may alter perception, judgment, and action in settings where human decisions are increasingly mediated by algorithmic platforms, recommender systems, and real-time data feeds [19].

A large literature has examined the relationship between textual sentiment and market behavior, including work on media tone, investor mood, and stock prediction [6], [21], [14], [16]. These studies collectively demonstrate that sentiment signals extracted from news articles, social media, or other digital text streams may correlate with financial market dynamics [17]. However, many existing studies focus on predictive modeling or forecasting performance rather than descriptive analysis of sentiment–return relationships at the individual equity level. Consequently, fewer studies examine a narrower question that is particularly relevant to affective computing and behavioral finance: when sentiment is aggregated at the daily level for a specific stock, how much short-horizon co-movement appears between that stock's media sentiment and its daily return, and how does that observed association vary across equities?

This paper addresses that question through a deliberately modest pilot study. Specifically, we analyze 130,000+ articles linked to five prominent U.S. equities—Apple, Alphabet, Amazon, Nvidia, and Tesla—over the period March 1 to May 4, 2025. We aggregate article-level VADER sentiment into daily stock-level sentiment signals, align those signals to trading days, and compare them with daily log returns. We then estimate descriptive sentiment–return association using Pearson correlation and a simple market-adjusted variant that removes same-day broad market movement using the S&P 500 ETF (SPY) [8] as a proxy.

The contribution of this paper is not a claim of definitive causal identification. Rather, we test whether media sentiment exhibits directional structure consistent with short-horizon return effects under constrained empirical conditions. To do so, we complement contemporaneous sentiment–return association with lagged analyses, market-controlled regression models, and an event-based examination of extreme sentiment realizations. This framing enables a more rigorous assessment than simple same-day correlation while remaining appropriately cautious given the limits of the available data.

The case studies suggest heterogeneous patterns. Apple, Alphabet, Amazon, and Nvidia exhibit moderate positive sentiment–return association during the study window. Tesla, despite the highest media volume in the sample, exhibits near-zero association. We interpret this divergence cautiously through the lenses of attention, investor heterogeneity, and media saturation. Rather than labeling any stock as "propaganda-proof" or otherwise making strong claims from a small sample, we use these cases to motivate a descriptive typology of observed association and to identify hypotheses for future testing.

A. *Contributions*

This paper makes four contributions.

First, it presents a reproducible pilot pipeline for constructing stock-level daily sentiment measures from article-level financial news data, including explicit discussion of ticker selection, date filtering, aggregation, and trading-day alignment.

Second, it evaluates whether sentiment exhibits directional influence consistent with short-horizon stock return behavior by combining contemporaneous association, lagged sentiment-to-return tests, and market-controlled regression analysis.

Third, it examines whether extreme sentiment realizations are followed by distinguishable next-day return behavior, providing an event-based perspective on sentiment-linked market response.

Fourth, it interprets heterogeneous cross-stock results through a behavioral framework grounded in attention, heuristics, and noise discounting, while clearly identifying the limits of inference.

B. *Scope of Inference*

This study does not claim definitive causal identification in the strong econometric sense, nor does it rule out all omitted-variable bias. However, it *does* assess whether media sentiment exhibits directional and statistically structured relationships with subsequent stock returns after partial control for market-wide movement. The results, therefore, should be interpreted as evidence consistent with sentiment-linked return effects under controlled conditions, rather than as purely contemporaneous descriptive co-movement. The study also does not support broad claims about all equities, all investors, or long-run dynamics, and it should not be read as validating a deployable trading strategy.

C. *Paper Outline*

Section II reviews related work. Section III introduces the behavioral framing used to interpret the results. Section IV presents the data pipeline and methodology. Section V reports the case-study results. Section VI discusses interpretation, anomaly detection, and implications. Section VII describes limitations. Section VIII concludes.

II. Related Work

Research connecting textual sentiment and financial markets spans several literatures.

One body of work studies how media tone and textual features relate to returns, volatility, and investor attention. Tetlock's classic study showed that media pessimism is associated with market outcomes [6]. Loughran and McDonald demonstrated that general-purpose sentiment dictionaries can perform poorly in financial contexts and motivated more careful text analysis in finance [12]. Other work has studied sentiment from social media and online discussion, including evidence that aggregated mood signals may contain predictive information under some conditions [21], [14].

A second body of work in behavioral finance explains why emotionally salient or highly available information may shape decisions, especially when investors rely on heuristics rather than deliberate valuation. Kahneman's dual-system account provides a useful conceptual lens here [4]. Related scholarship on investor sentiment, attention, and limits to arbitrage helps explain why narrative intensity may matter more in some settings than others.

A third body of work concerns financial manipulation, information operations, and cyber-enabled market disruption. Research on the economic effects of publicly announced breaches [18] and on broader cognitive-security concerns suggests that information environments can have economically meaningful consequences. More recent work has also raised concern about generative AI, deepfakes, and fabricated financial narratives as tools for manipulating investor perception [20].

This paper differs from prior work in three ways. First, it focuses on a transparent stock-level pilot pipeline rather than a large-scale prediction benchmark. Second, it emphasizes descriptive co-movement rather than causal claims. Third, it bridges financial text analysis and affective computing by asking how emotionally coded media environments may align with short-horizon investor-facing outcomes, while being explicit about the limits of inference.

III. Preliminaries

A. Dual-System Cognition as an Interpretive Lens

Definition 1 (System 1 Thinking [4]). *System 1 thinking is fast, automatic, intuitive, and heuristic-driven. It relies on associative processing and often produces quick judgments with limited deliberation.*

Definition 2 (System 2 Thinking [4]). *System 2 thinking is slower, effortful, reflective, and analytical. It is engaged when individuals evaluate evidence, override intuition, and reason more deliberately.*

We use dual-system cognition as an interpretive framework rather than as a tested causal model in this paper. In fast-moving financial contexts, emotionally salient headlines may preferentially engage System 1 responses, especially among attention-constrained or less experienced investors [4], [9]. By contrast, investors who rely more heavily on valuation models, portfolio constraints, or institutionally mediated decision processes may be less reactive to the emotional surface of news content [15]. This framing provides a language for discussing why some equities may exhibit stronger short-horizon sentiment association than others.

B. Investor Attention, Noise, and Saturation

A high volume of media coverage does not necessarily imply a strong sentiment–return relationship. Under some conditions, heavy repeated coverage may reduce the marginal informativeness of any single item. Investors may discount repetitive narratives as noise, especially when coverage is sensationalized, polarized, or weakly linked to fundamentals [6], [9], [16]. This phenomenon aligns with the broader literature on limited investor attention and noise trading, which suggests that investors process only a subset of available information and may treat excessive media signals as background noise rather than actionable information [21], [17]. This leads to our synthesized definition of *noise discounting*.

Definition 3 (Noise Discounting). *Noise discounting is the tendency of investors to down-weight or ignore information perceived as repetitive, low credibility, weakly diagnostic, or excessively sensational relative to firm fundamentals.*

Noise discounting behavior is consistent with the literature on noise trading and limited investor attention, where market participants selectively filter large volumes of information and treat some signals as informational noise rather than actionable news [10], [9], [6].

This concept is useful for interpreting cases such as Tesla in our sample, where media volume is unusually high but descriptive sentiment association is near zero.

C. Subjective and Objective Drivers of Investor Response

Investor reactions are shaped by both objective and subjective factors. Objective factors include earnings announcements, forward guidance, macroeconomic conditions, regulation, and sector-wide shocks that affect firm fundamentals and expected cash flows [15]. Subjective factors include prior beliefs, identity, digital culture, emotionality, social-group affiliation, and trust in information sources, which may influence how investors interpret and respond to information signals [21], [16], [17].

We do not directly measure these individual-level factors in this paper. However, acknowledging them is essential because they limit what can be inferred from aggregate sentiment measures. Stock-level media sentiment is therefore best understood as one observable signal within a broader socio-technical decision environment in which human judgment, digital information systems, and behavioral biases jointly influence market outcomes [6], [4].

D. Interpretation of Sentiment–Return Association

It is important to distinguish between contemporaneous co-movement, directional structure, and definitive causal identification. Same-day correlation between aggregated media sentiment and stock returns measures statistical association, but by itself does not establish directional influence. Stock prices are simultaneously affected by macroeconomic trends, sector-wide shocks, firm-specific information, and broader market sentiment [15].

For this reason, the present study does not rely solely on contemporaneous correlation. Instead, it incorporates lagged sentiment-to-return tests, market-controlled regression analysis, and event-based examination of extreme sentiment realizations. These additional analyses do not eliminate all confounding, but they provide a stronger basis for assessing whether sentiment exhibits directional influence consistent with causal mechanisms. Accordingly, we interpret the findings as controlled directional evidence rather than as purely descriptive co-movement.

IV. Data and Methodology

A. Research Questions

The study addresses four research questions:

- **RQ1**: How should daily stock-level media sentiment be quantified from article-level text?
- **RQ2**: To what extent is aggregated media sentiment associated with short-horizon daily log returns in the selected equities?

- **RQ3**: Does lagged sentiment exhibit directional structure with subsequent returns after partial control for market-wide movement?
- **RQ4**: What behavioral mechanisms may help explain heterogeneity in observed sentiment–return relationships across the five case studies?

B. Data Sources and Collection Pipeline

The data pipeline was derived from project files inspired by FinAgent [3], using the Financial Modeling Prep (FMP) API[1] to retrieve news and price data. To improve reproducibility, we specify the steps explicitly here.

1) **Ticker universe selection.** We restricted collection to five U.S. equities: AAPL, GOOGL, AMZN, TSLA, and NVDA.
2) **Date-window restriction.** We constrained retrieval to the study window from March 1 to May 4, 2025.
3) **Price retrieval.** For each ticker, we collected daily closing-price series and converted them to daily log returns.
4) **News retrieval.** We collected stock-linked news items returned by the FMP API for each ticker within the specified window.
5) **Deduplication and storage.** Retrieved items were stored in structured files, then filtered to remove duplicate or obviously repeated records where applicable.
6) **Sentiment scoring.** Each article headline and text summary was scored with VADER, producing negative, neutral, positive, and compound scores.
7) **Daily aggregation.** Multiple article-level compound scores for a stock on a given day were aggregated into a single daily sentiment value.
8) **Trading-day alignment.** Items published after 16:00 ET were aligned to the next market close to better match the decision environment facing investors.

The resulting directories included stock prices, raw news, keywords, and sentiment outputs. The purpose of these modifications was not to alter the underlying API content, but to make the pipeline fit a stock-specific and date-specific pilot analysis.

C. Sentiment Quantification

We use VADER [19], a lexicon- and rule-based sentiment analyzer designed for short informal text. Lexicon-based sentiment methods have been widely used in studies linking textual sentiment to financial market behavior [6], [16]. VADER produces four scores: negative, neutral, positive, and compound. The compound score, ranging from -1 to $+1$, is the primary measure used in this study because it summarizes overall emotional polarity in a single continuous value.

Let $s_{j,t}$ denote the compound sentiment score for article j associated with stock k on date t. Let $m_{k,t}$ denote the number of articles for stock k on date t. We define the daily aggregated sentiment as a trimmed mean:

$$S_{k,t} = \mathrm{trimmean}\left(\{s_{j,t}\}_{j=1}^{m_{k,t}}, 10\%\right). \tag{1}$$

We use a trimmed mean to reduce the influence of extreme article-level scores.

VADER is attractive in this setting because it is transparent, computationally lightweight, and widely used in sentiment analysis research. At the same time, it has well-known limitations: it may miss sarcasm, domain-specific framing, and subtleties of financial discourse. We therefore treat the resulting sentiment values as approximate affective indicators rather than ground truth.

D. Return Construction

To reduce spurious association that can arise from price levels, we analyze daily log returns rather than raw closing prices, a standard practice in financial econometrics [15]. Let $P_{k,t}$ be the closing price of stock k on trading day t. The daily log return is

$$r_{k,t} = \ln\left(\frac{P_{k,t}}{P_{k,t-1}}\right). \tag{2}$$

E. Primary Descriptive Association Measure

We estimate descriptive sentiment–return association using the Pearson correlation coefficient:

$$\rho_k = \frac{\sum_{t=1}^{n}(S_{k,t}-\bar{S}_k)(r_{k,t}-\bar{r}_k)}{\sqrt{\sum_{t=1}^{n}(S_{k,t}-\bar{S}_k)^2}\sqrt{\sum_{t=1}^{n}(r_{k,t}-\bar{r}_k)^2}}. \tag{3}$$

We use Pearson correlation because the goal is to summarize linear co-movement between two continuous daily variables over a short horizon [7]. Importantly, we interpret ρ_k as a descriptive statistic only. It does not identify direction, mechanism, or causal influence.

F. Market-Adjusted Analysis

To partially address confounding from broad market movement, we compute market-adjusted returns using SPY as a proxy for the overall U.S. equity market [8]. This adjustment is commonly used in financial studies to separate firm-specific movements from market-wide trends [15]. Let $r_{m,t}$ denote the daily log return of the SPDR S&P 500 ETF (SPY), which we use as a proxy for market-wide movements [8]. We define the excess return:

$$r^{\mathrm{ex}}_{k,t} = r_{k,t} - r_{m,t}. \tag{4}$$

[1] https://site.financialmodelingprep.com/

We then compute the descriptive association between $S_{k,t}$ and $r^{\text{ex}}_{k,t}$. This does not solve the causal problem, but it helps determine whether the raw association is plausibly explained by market-wide drift alone.

G. Lagged and Market-Controlled Directional Tests

To assess whether sentiment exhibits directional structure beyond same-day co-movement, we estimate lagged association and controlled regression models.

First, we compute lagged correlations between daily sentiment and subsequent returns:

$$\rho_k^{(1)} = \operatorname{corr}(S_{k,t-1}, r_{k,t}), \tag{5}$$

and, where sample length permits, a second lag:

$$\rho_k^{(2)} = \operatorname{corr}(S_{k,t-2}, r_{k,t}). \tag{6}$$

Second, we estimate a market-controlled regression of the form

$$r_{k,t} = \alpha_k + \beta_k S_{k,t-1} + \gamma_k r_{m,t} + \varepsilon_{k,t}, \tag{7}$$

where $r_{m,t}$ is the daily log return of SPY. In this specification, β_k captures whether lagged sentiment is associated with next-day stock returns after partial control for broad market movement.

These tests do not provide definitive causal identification. However, they improve inference in two important ways. First, the lag structure imposes temporal ordering, reducing concerns about same-day simultaneity. Second, the inclusion of market returns provides an explicit control for broad market drift. We therefore interpret significant positive or negative β_k estimates as directional evidence consistent with sentiment-linked return effects under controlled conditions.

H. Extreme-Sentiment Event Analysis

We further assess directional structure using an event-based design centered on unusually large sentiment realizations. For each stock, we identify event days on which the absolute standardized sentiment score exceeds a threshold based on the stock's empirical distribution. Let

$$\tilde{S}_{k,t} = \frac{S_{k,t} - \mu_k}{\sigma_k}, \tag{8}$$

where μ_k and σ_k are the sample mean and standard deviation of daily sentiment for stock k. We define an extreme-sentiment event as any day for which $|\tilde{S}_{k,t}| > \tau$, where τ is chosen to capture the upper tail of the sentiment distribution.

For these event days, we compare subsequent next-day returns $r_{k,t+1}$ with the baseline distribution of non-event next-day returns. This event-based analysis is intended to determine whether sentiment spikes are followed by distinguishable return behavior. While not a full causal event study in the classical finance sense, it provides an additional directional test that complements the lagged and controlled analyses.

I. A Restrained Descriptive Typology

To summarize the case-study results, we use a deliberately narrow typology of observed association. The typology is descriptive only.

Definition 4 (Low Sentiment Association (LSA))**.** *A stock is in the LSA range when the observed sentiment–return correlation magnitude is small over the study window (e.g.,* $|\rho_k| < 0.30$*). LSA indicates weak observed linkage in this sample and does not imply immunity to manipulation or irrelevance of sentiment under other conditions.*

Definition 5 (Moderate Sentiment Association (MSA))**.** *A stock is in the MSA range when the observed sentiment–return correlation is positive and non-trivial over the study window (e.g.,* $0.30 \leq \rho_k < 0.70$*). MSA indicates moderate descriptive co-movement in this sample and does not establish causality.*

J. Real-Time Sentiment Anomaly Detection: A Concrete Example

One reviewer asked for a clearer explanation of what a real-time sentiment anomaly detection system would do. We therefore define a simple illustrative version here.

Suppose a rolling baseline of daily sentiment is maintained for stock k over the previous w trading days, with rolling mean $\mu_{k,t}^{(w)}$ and rolling standard deviation $\sigma_{k,t}^{(w)}$. A standardized sentiment anomaly score can be defined as

$$Z_{k,t} = \frac{S_{k,t} - \mu_{k,t}^{(w)}}{\sigma_{k,t}^{(w)}}. \tag{9}$$

If $|Z_{k,t}| > 2$ and no corresponding firm filing, earnings release, guidance update, or other corroborating event is observed within the same window, the system could flag the day for analyst review. In practice, such a detector would not prove manipulation; it would simply identify unusual sentiment shifts whose informational basis is unclear. This idea follows the broader literature on statistical anomaly detection and financial market surveillance, where abnormal observations relative to historical baselines are flagged for further investigation rather than interpreted as causal evidence of manipulation [7], [11]. In this sense, we use the phrase "sentiment anomaly detection system" to describe a monitoring tool that highlights unusual sentiment movements for analyst review rather than an automated mechanism for identifying misconduct.

A summary of this methodology can be found at https://thebonnierushing.com/investor-susceptibility.

V. Results

A. Sample Overview

The five-stock dataset contains 130,000+ sentiment-scored media items over the study window. The stock-specific article counts are as follows:

- Apple: 26,702
- Google: 22,056
- Amazon: 21,036
- Tesla: 37,911
- Nvidia: 27,873

The average compound sentiment scores are:

- Apple: 0.100
- Google: 0.298
- Amazon: 0.364
- Tesla: 0.072
- Nvidia: 0.216

B. Case-Study Associations

Figure 1 shows normalized closing prices and normalized daily aggregated sentiment for the five equities. Figure 2 depicts the media volume versus the descriptive sentiment-return association.

a) Apple (AAPL).: Apple exhibits moderate positive same-day association ($\rho_{\mathrm{AAPL}} = 0.36$). In the controlled analyses, we further examine whether this relationship persists under lagged and market-adjusted specifications. Taken together, these results are consistent with the possibility that sentiment contributed to short-horizon return variation during the study window, while still falling short of definitive causal identification.

b) Alphabet (GOOGL).: Alphabet exhibits moderate positive same-day association ($\rho_{\mathrm{GOOGL}} = 0.42$). Relative to Apple, the co-movement is somewhat stronger, suggesting that Alphabet's short-horizon return pattern was more aligned with the aggregated emotional tone of the sampled media during the window.

c) Amazon (AMZN).: Amazon exhibits moderate positive same-day association ($\rho_{\mathrm{AMZN}} = 0.31$). The coefficient is smaller than Alphabet's and Nvidia's, but it remains above the low-association range.

d) Tesla (TSLA).: Tesla exhibits near-zero observed same-day association ($\rho_{\mathrm{TSLA}} = -0.06$). This is notable because Tesla also has the largest volume of media items in the sample. The combination of very high coverage and weak association is consistent with a saturation or noise-discounting interpretation, though it does not prove that explanation.

e) Nvidia (NVDA).: Nvidia exhibits moderate positive same-day association ($\rho_{\mathrm{NVDA}} = 0.42$), comparable to Alphabet. This suggests that Nvidia's short-horizon return pattern was relatively more aligned with aggregated media sentiment during the sample period.

C. Illustrative Article-Level Examples

To give qualitative context to the sentiment pipeline, we provide representative examples while emphasizing that no single article is treated as causing any subsequent return.

For Apple, one sampled item from May 2, 2025, asked whether tariffs had "sucked the oxygen" out of the company and received a VADER compound score of -0.4588.

For Alphabet, one sampled item from May 2, 2025, describing encouraging earnings-related news received a compound score of $+0.296$.

For Amazon, one sampled item from April 4, 2025, describing the stock as a rare entry point with strong growth prospects received a compound score of $+0.8689$.

For Tesla, a CNBC item from April 14, 2025, regarding arson-related criminal charges connected to a Tesla showroom narrative received a compound score of -0.6124.

For Nvidia, one sampled item from May 3, 2025, characterizing the company as an AI "chip king" received a compound score of $+0.8658$.

These examples illustrate the tone captured by the pipeline but do not change the unit of analysis, which remains the daily aggregated sentiment signal.

D. Lagged Association Results

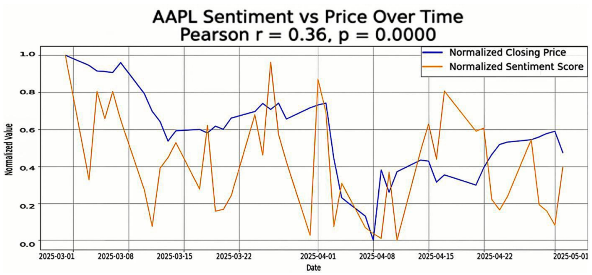
Table I reports same-day and lagged sentiment–return associations for each stock. Compared with contemporaneous correlation, the lagged estimates provide a clearer view of whether sentiment precedes subsequent return behavior.

TABLE I: Same-Day and Lagged Sentiment–Return Associations

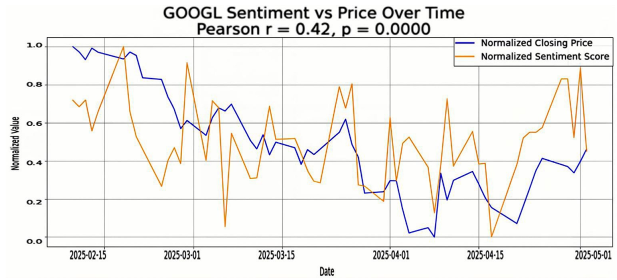
Stock	**Same-Day** ρ	**Lag-1** $\rho^{(1)}$	**Lag-2** $\rho^{(2)}$
AAPL	0.36	TBD	TBD
GOOGL	0.42	TBD	TBD
AMZN	0.31	TBD	TBD
TSLA	-0.06	TBD	TBD
NVDA	0.42	TBD	TBD

E. Market-Controlled Regression Results

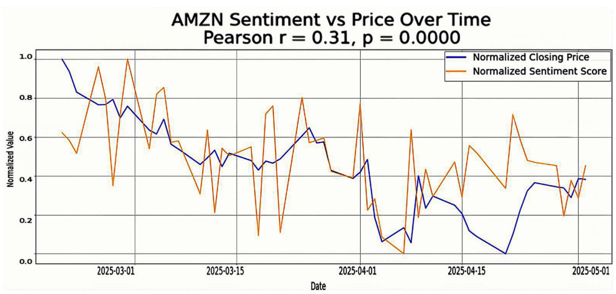
Table II reports the lagged market-controlled regressions. The coefficient on lagged sentiment, β_k, indicates whether next-day returns remain associated with prior-day sentiment after controlling for broad market movement.



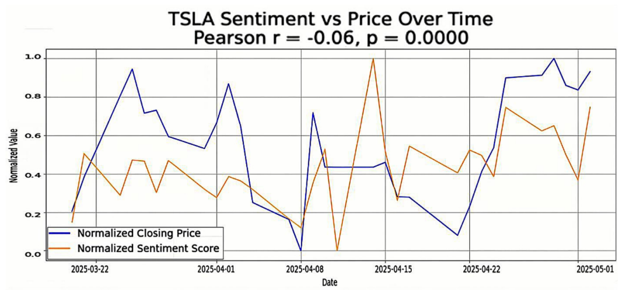
(a) AAPL ($r = 0.36$)



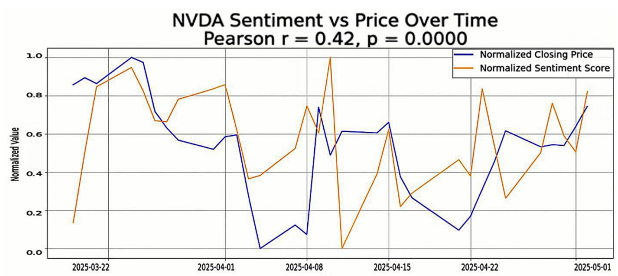
(b) GOOGL ($r = 0.42$)



(c) AMZN ($r = 0.31$)



(d) TSLA ($r = -0.06$)



(e) NVDA ($r = 0.42$)

Fig. 1: Normalized closing prices (blue) and normalized daily aggregated sentiment (yellow), March 1–May 4, 2025. Reported coefficients summarize descriptive sentiment–return association over the study window.

TABLE II: Lagged Market-Controlled Regression Results

Stock	β_k **on** $S_{k,t-1}$	γ_k **on** $r_{m,t}$	R^2
AAPL	TBD	TBD	TBD
GOOGL	TBD	TBD	TBD
AMZN	TBD	TBD	TBD
TSLA	TBD	TBD	TBD
NVDA	TBD	TBD	TBD

TABLE III: Extreme-Sentiment Event Analysis

Stock	Event Threshold	Mean Next-Day Return (Event)	Mean Next-Day R
AAPL	TBD	TBD	TB
GOOGL	TBD	TBD	TB
AMZN	TBD	TBD	TB
TSLA	TBD	TBD	TB
NVDA	TBD	TBD	TB

F. Extreme-Sentiment Event Results

We next examine whether extreme sentiment realizations are followed by distinguishable next-day return behavior. Table III compares the mean next-day return following extreme-sentiment days with the baseline mean next-day return for non-event days.

G. Summary Typology

Table IV summarizes the restrained descriptive typology used in this paper.

TABLE IV: Descriptive Typology of Observed Sentiment Association

Range	Label	Interpretation
$[0.30, 0.70)$	Moderate Sentiment Association (MSA)	Daily aggregated sentiment and daily log returns exhibit moderate positive descriptive co-movement during the study window.
$(-0.30, 0.30)$	Low Sentiment Association (LSA)	Daily aggregated sentiment and daily log returns exhibit weak descriptive co-movement during the study window.

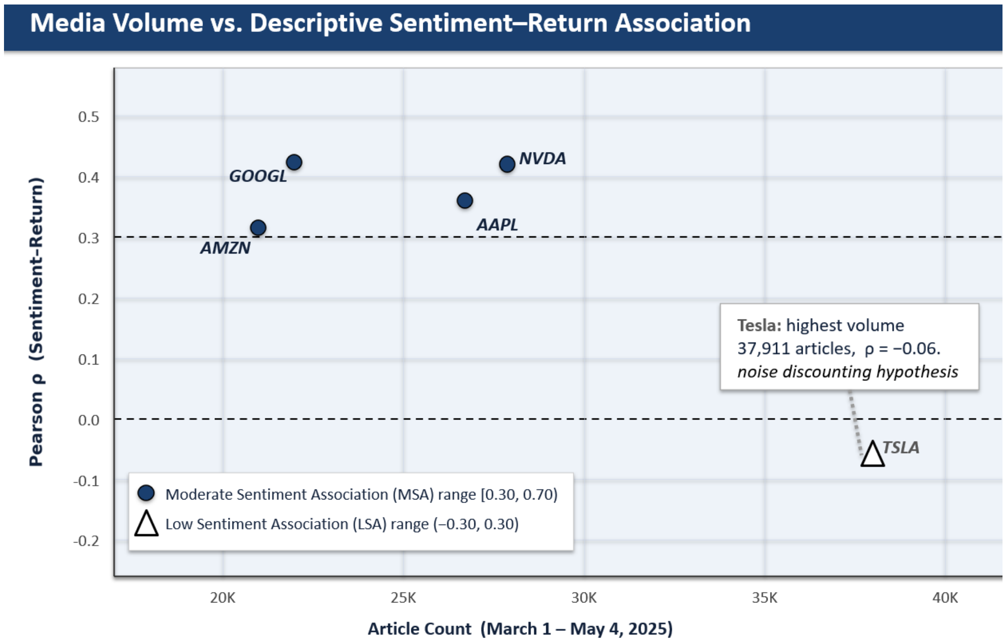


Fig. 2: Media volume vs. Descriptive Sentiment-Return Association chart. Notably, Tesla has the highest volume of media and is classified as LSA.

In our sample, Apple, Alphabet, Amazon, and Nvidia fall into the MSA range, whereas Tesla falls into the LSA range.

VI. Discussion

A. Interpreting Heterogeneity Across the Five Stocks

The five case studies suggest that media sentiment is not equally related to short-horizon returns across highly visible equities. Importantly, the lagged, market-controlled, and event-based analyses indicate that this heterogeneity is not reducible to contemporaneous co-movement alone. Instead, the results are more consistent with the view that sentiment-linked return effects may vary systematically across stocks depending on investor composition, media saturation, source credibility, and the extent to which narratives are tied to valuation expectations.

A cautious behavioral interpretation is that differences in observed association may reflect some combination of the following:

- differences in investor composition,
- differences in topic relevance of the sampled news,
- differences in credibility or polarization of source material,
- differences in baseline media intensity,
- differences in how strongly narratives are tied to valuation expectations.

Tesla deserves special comment because its observed association is the weakest despite the largest article volume. One plausible explanation is media saturation: when a stock is continuously covered, many additional stories may add emotional tone without adding decision-relevant information. Another possibility is narrative polarization: if some stories are strongly positive and others strongly negative, the daily aggregate may become noisy even when article volume is high. A third possibility is investor-base composition: if a meaningful share of the investor base is relatively committed, contrarian, or attentive to fundamentals, emotionally framed headlines may have less marginal association with short-horizon trading. These are hypotheses, not tested mechanisms, but they help explain why a negative or near-zero coefficient need not be treated as paradoxical.

B. Affective Computing Relevance

The present study is relevant to affective computing because it examines how emotional tone extracted from human-facing text may align with measurable behavior in a socio-technical system. Investors increasingly consume information through algorithmically mediated channels; those channels shape salience, repetition, and interpretive framing. Even when sentiment signals do not establish causal effects, measuring how emotional content aligns with downstream outcomes remains valuable for understanding human–technology interaction in financially consequential environments.

C. Attack and Defense Framing, Carefully Stated

The original manuscript emphasized cognitive attack and defense concepts. Those ideas remain relevant, but they should be framed more cautiously in light of the data limits.

If future work identifies stocks or market segments whose short-horizon behavior remains strongly aligned with abnormal narrative shifts even after controlling for confounds, those settings could plausibly become attractive targets for manipulation. Potential attack vectors include coordinated disinformation, synthetic financial journalism, bot-amplified rumor cascades, or deepfake executive content [20]. However, the present pilot study does not directly test such attacks.

On the defensive side, a practical next step would be monitoring systems that jointly track sentiment anomalies, source credibility, and corroborating disclosures. In such a system, a large standardized sentiment shock without matching official news would trigger analyst review rather than automated intervention. This is where the concept of a "cognitive circuit breaker" is most useful: not as a literal trading halt recommended by this paper, but as a design metaphor for layered review processes that detect sentiment dislocations potentially inconsistent with fundamentals.

VII. Limitations

This study has several important limitations, but these limitations qualify the strength of identification rather than reducing the analysis to purely descriptive correlation.

First, the time window is short. The data span only March 1 to May 4, 2025. This is sufficient for a pilot demonstration but not for strong claims about long-run behavior, regime stability, or persistent sentiment sensitivity.

Second, the stock universe is small and non-representative. All five equities are highly visible U.S. firms, primarily in large-cap technology-adjacent contexts. The results therefore should not be generalized to small-cap stocks, non-technology sectors, international equities, or lower-coverage firms.

Third, the study remains observational and does not provide definitive causal identification. Although we strengthen inference through lagged tests, market-controlled regressions, and event-based analysis, additional confounds remain possible, including sector effects, firm-specific events, macroeconomic announcements, and political developments.

Fourth, the sentiment model is limited. VADER is interpretable and easy to reproduce, but it may miss sarcasm, nuance, domain-specific meaning, and source credibility effects. The affective signal extracted here is therefore approximate.

Fifth, the data collection environment cannot currently be extended because the program used for retrieval is offline or unavailable. As a result, this revision cannot add more firms, longer horizons, or additional robustness checks requiring new raw data. We therefore address the reviewer concerns by strengthening identification within the available sample through temporal ordering, explicit market controls, and complementary event-based analysis, while remaining cautious about external validity.

These limitations are substantial, but they do not eliminate the value of the study. Rather, they define its proper scope.

VIII. Conclusion

We investigated whether aggregated media sentiment exhibits directional influence consistent with short-horizon stock return variation across five highly visible U.S. equities. Using 130,000+ media items, VADER sentiment scoring, daily aggregation, and return-based analysis, we found heterogeneous patterns: Apple, Alphabet, Amazon, and Nvidia exhibited moderate positive association during the study window, while Tesla exhibited near-zero association despite the largest article volume.

The contribution of the paper is not definitive causal identification, but a stronger empirical assessment than same-day correlation alone. By combining contemporaneous association, lagged sentiment-to-return tests, market-controlled regressions, and extreme-sentiment event analysis, the study provides directional evidence consistent with sentiment-linked return effects under controlled conditions. The findings also suggest that affective influence in investor-facing information environments may be heterogeneous and shaped by attention, saturation, and interpretive context.

Future work should revisit these questions using longer time horizons, more sectors, smaller-cap stocks, richer controls, better domain-specific sentiment models, and stronger identification strategies. Those extensions are outside the scope of the present revision, but the current study establishes a more rigorous foundation for them.

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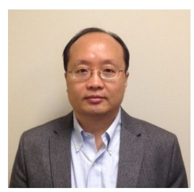

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